

Italia Rodriguez

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Engineering graduate with hands-on experience in product development, manufacturing, quality, and production operations. Seeking a challenging opportunity that leverages my technical versatility, product development experience, and ability to adapt quickly across disciplines.

EDUCATION

The University of Texas at Austin

May 2026

B.S., Biomedical Engineering

Austin, TX

- **GPA: 3.7/4.0**
- **Study Abroad:** Micro- and Nanotechnology for Sensors, Kyoto, Japan (May 2025)

EXPERIENCE

Detekt Biomedical

February 2024 – Present

Product Design and Development Manager | June, 2025 – Present

Austin, TX

- Designed and prototyped 15+ customer-driven products and accessories in SolidWorks to expand platform functionality and support new customer applications.
- Developed customer-specific product adaptations, including injection-molded components, using DFM principles to improve compatibility across multiple cartridge formats and support international partners.
- Produced manufacturing drawings and engineering documentation to streamline production, accelerate design iteration, and facilitate cross-functional collaboration.
- Leveraged hands-on manufacturing and quality experience to develop practical, manufacturable engineering solutions that balanced customer requirements with production needs.

Lead Assembly Technician | September, 2024 – June, 2025

- Led and trained three production technicians while coordinating daily manufacturing operations to consistently achieve monthly production targets (~250 devices/month).
- Designed tooling and implemented process improvements that reduced assembly errors by approximately 20%, improving manufacturing efficiency and production repeatability.
- Supported production planning, inventory management, quality reporting, and SOP development to improve workflow consistency and operational efficiency.

Assembly and QC Technician | Feb, 2024 – September, 2024

- Assembled, calibrated, and tested diagnostic instrumentation while maintaining quality standards and supporting on-time production schedules.
- Performed quality inspections and troubleshooting to verify product performance, identify defects, and ensure reliable customer delivery.

PROJECTS

3D Fetal Heart Education Platform - Senior Design Project

- Collaborated with a multidisciplinary engineering team and industry sponsor to develop an interactive 3D fetal heart education platform for congenital heart disease visualization.
- Translated clinical requirements into an accessible educational tool through iterative design and model evaluation.

Wearable UV Sensor & Mobile App

- Developed a wearable UV monitoring system integrating Arduino, Bluetooth Low Energy (BLE), and a companion mobile application to deliver real-time exposure alerts.
- Contributed to embedded systems integration, usability testing, and iterative design improvements to enhance user experience.

SKILLS

- **Engineering:** Product Development, DFM, Injection Mold Design, 3D Printing, Rapid Prototyping, Process Improvement, Quality Assurance
- **Software:** SolidWorks, MATLAB, Python, Arduino, Microsoft Excel
- **Languages & Interests:** Spanish (Fluent), Japanese (Beginner); Robotics, Miniature Electronics, Language Learning